



THE KENYA NATIONAL EXAMINATIONS COUNCIL
Kenya Certificate of Secondary Education



443/1 -

AGRICULTURE

- Paper 1

Nov. 2017 - 2 hours

Name Index Number

Candidate's Signature Date

Instructions to candidates

- Write your name and index number in the spaces provided above.
- Sign and write the date of examination in the spaces provided above.
- This paper consists of **three** sections; **A, B and C**.
- Answer **all** the questions in section **A** and **B**.
- Answer any **two** questions in section **C**.
- All answers should be written in the spaces provided.
- This paper consists of 16 printed pages.**
- Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.**
- Candidates should answer all the questions in English.**

For Examiner's Use Only

Section	Question	Maximum Score	Candidate's Score
A	1-15	30	
B	16-19	20	
C		20	
		20	
Total Score		90	



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443/1

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SECTION A (30 marks)

*Answer **all** the questions in this section in the spaces provided.*

1. Give **four** reasons for intercropping in crop production. (2 marks)
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.....
2. State **four** characteristics of extensive farming system. (2 marks)
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3. State **four** minimum tillage practices. (2 marks)
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.....
4. Give **four** reasons why burning of land is discouraged. (2 marks)
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5. State **four** functions of Young Farmers' Clubs. (2 marks)

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6. Distinguish between a perfect and an imperfect market. (1 mark)

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7. State **four** reasons for practising intensive hedgerow agroforestry. (2 marks)

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8. State **four** management practices in intensive hedgerow agroforestry. (2 marks)

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9. State the meaning of each of the following terms as used in crop production:

(a) crop rotation

(1 mark)

.....

.....

(b) pruning

(1 mark)

.....

.....

(c) rogueing.

(1 mark)

.....

.....

10. State **four** factors that determine the depth of planting.

(2 marks)

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11. State **four** disadvantages of broadcasting seeds during planting.

(2 marks)

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12. State **four** characteristics of a good site for a nursery bed. (2 marks)

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13. Name **four** types of market structures in agricultural marketing. (2 marks)

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14. Name **one** crop that is propagated by each of the following:

- (a) stem tuber (½ mark)

.....

- (b) split (½ mark)

.....

- (c) slip (½ mark)

.....

- (d) bulbil. (½ mark)

.....

15. State **four** advantages of using certified seeds. (2 marks)

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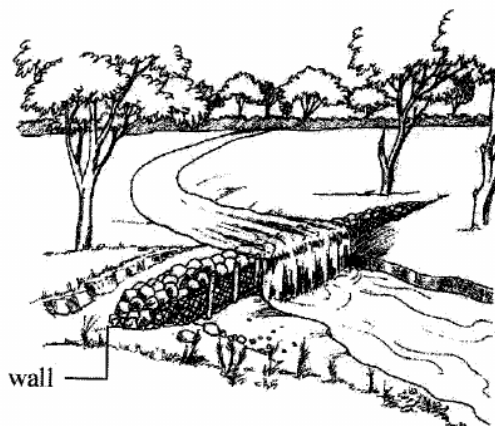
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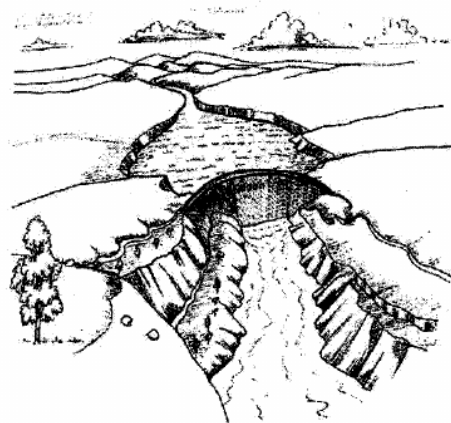
SECTION B (20 marks)

*Answer **all** the questions in this section in the space provided.*

16. The diagram below illustrates water storage structures.



E



F

- (a) Identify the structure labelled E. (1 mark)

.....

- (b) Give **two** reasons why farmers prefer structure F to E. (2 marks)

.....

.....

- (c) State **two** maintenance practices for the structure labelled F. (2 marks)

.....

.....



17. A farmer has a piece of land on which he can grow maize, cabbages and beans. The expected yields and selling prices of the three crops are shown below.

Crop	Yield (kg)	Selling prices (Ksh/kg)
Maize	4000	40
Cabbages	2800	60
Beans	3000	80

If the cost of producing any of the three crops is the same

- (a) Which crop should the farmer grow? Show your working. (2 marks)

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- (b) (i) State the farmer's opportunity cost. (1 mark)

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- (ii) Give a reason for your answer in (b) (i) above. (1 mark)

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- (c) Give a reason why farmers always have to make a choice on the enterprise to implement on the farm. (1 mark)

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18. The following is a list of plant nutrients:

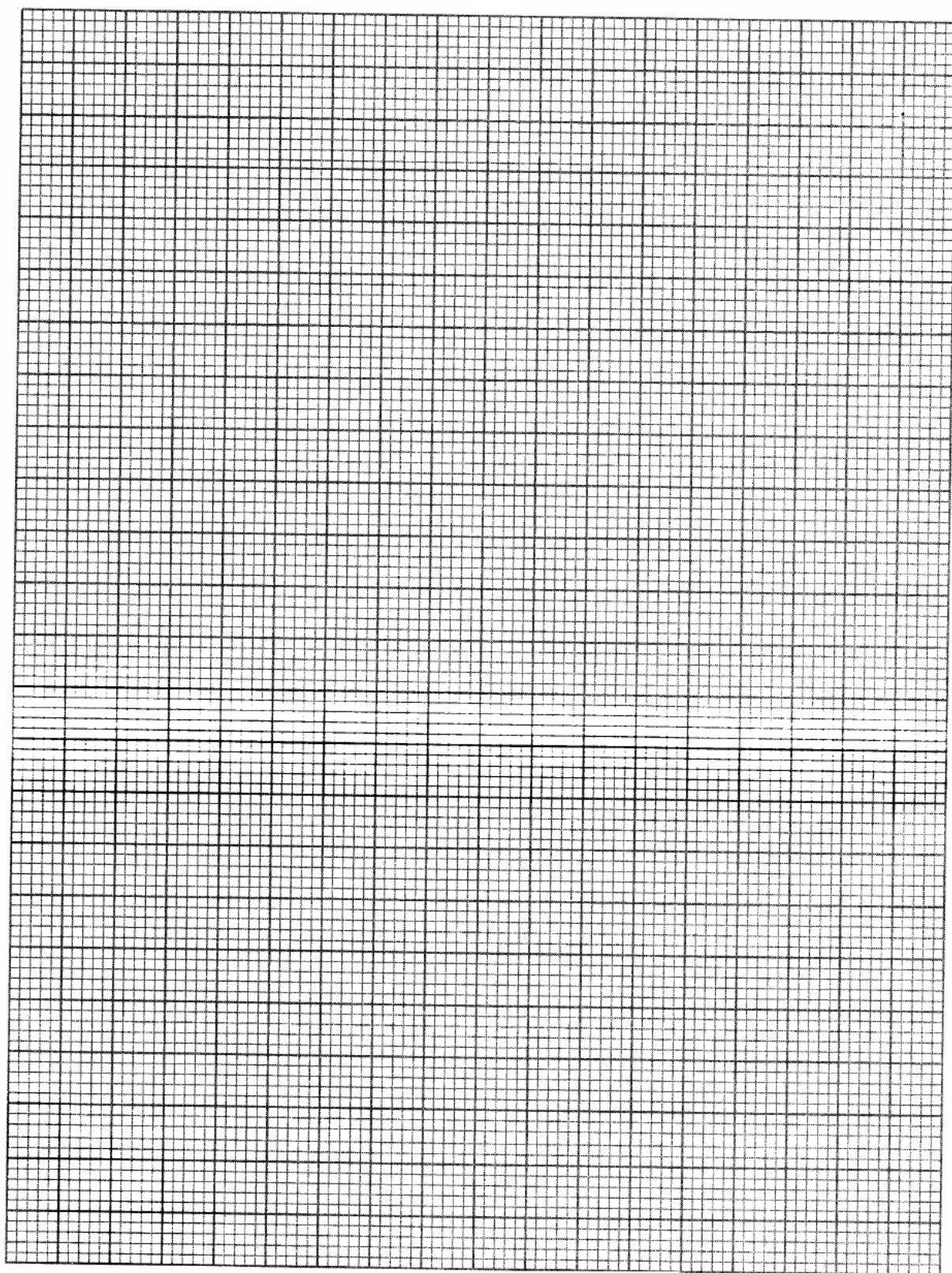
Copper, calcium, nitrogen, molybdenum, zinc, phosphorous, carbon, sulphur, iron and magnesium.

- (a) Which one of the above plant nutrients is mainly known for
- (i) promoting root development (1 mark)
.....
 - (ii) preventing blossom end rot disease (1 mark)
.....
 - (iii) Strengthening plant stalks to prevent lodging. (1 mark)
.....
- (b) Name **two** forms in which nitrogen is absorbed from the soil by plants. (2 marks)
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- This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



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443/1